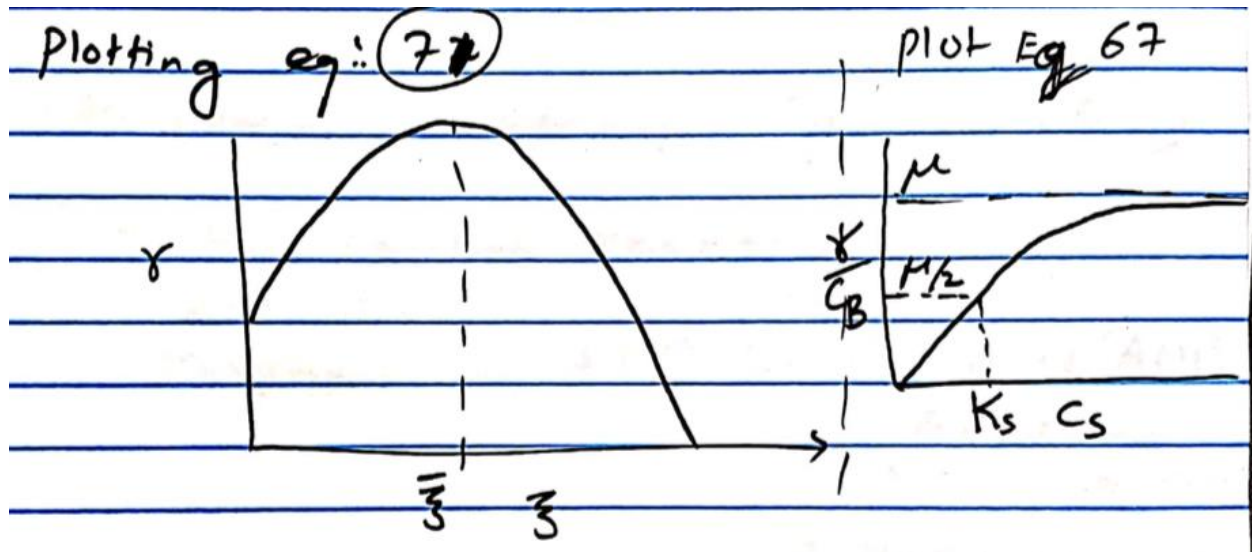


When $\frac{\partial r}{\partial \xi} = 0$, from (72)

$$C_B = \frac{1}{\alpha K_S} (K_S + C_S) C_S \quad (73)$$

Point of intersection of (69) and (73) represents critical value of ξ , below which the reaction is auto catalytic.



** 3. Energy Substrate Kinetics

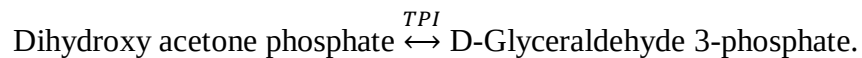
*Enzymes are biological molecules such as protein molecules that act as catalytic for biochemical reaction.

*Enzymes are highly specific catalysts that act on one or multiple substrates (reactant molecules) to form one or multiple products.

*Example:

1.Enzyme- Triose-Phosphate Isomerase (TPI).

*Biochemical reaction (Glycolysis):



TPI increases the rate by billion times. TPI deficiency can cause severe neurological disorder.

2.Enzyme- Adenyl Cyclase (AC)

*Biochemical reaction

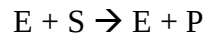
Adenosine Triphosphate (ATP) $\xrightleftharpoons{AC}$ Cyclic (AMP) Adenosine Monophosphate + Pyrophosphates.

3. Enzyme – Ribozymes (Ribonucleic acid enzymes)

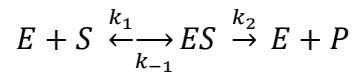
*Biochemical reaction: Cleavage or ligation of RNA, DNA and peptide bond.

****Mechanism of single substrate enzyme reaction:**

*Consider an enzyme (E) which can convert certain substrate into product (P)



Michaelis-Menton Kinetics views this reaction as occurring in following steps



$$\frac{dC_{ES}}{dt} = k_1 C_E C_S - k_{-1} C_{ES} - k_2 C_{ES} = k_1 C_E C_S - (k_{-1} + k_2) C_{ES} \quad (74)$$

Assume PSSH:

$$\begin{aligned} \frac{dC_{ES}}{dt} = 0 &= k_1 C_E C_S - (k_{-1} + k_2) C_{ES} \\ C_{ES} &= \frac{k_1 C_E C_S}{(k_{-1} + k_2)} \quad (75) \end{aligned}$$

Assume $E + S \rightleftharpoons ES$ such that $k_{-1} \gg k_2$ and $ES \rightarrow P + E$ is the rate determining step.

From (75)

$$C_{ES} = \frac{k_1 C_E C_S}{(k_{-1})} = K_1 C_E C_S \quad (76)$$

There is no net consumption of E. Therefore, total amount of E is always constant.

$$C_{E,o} = C_E + C_{ES} = C_E + K_1 C_E C_S = C_E (1 + K_1 C_S) \quad (77)$$

The rate of product formation:

$$r = k_2 C_{ES} = k_2 K_1 C_E C_S = \frac{k_2 K_1 C_S C_{E,o}}{1 + K_1 C_S} = \frac{r_{max} C_S}{K_S + C_S} \quad (78)$$

Where $r_{max} = k_2 C_{E,o}$ (maximum production rate) and $K_S = K_1^{-1}$ is Michaeli's constant.

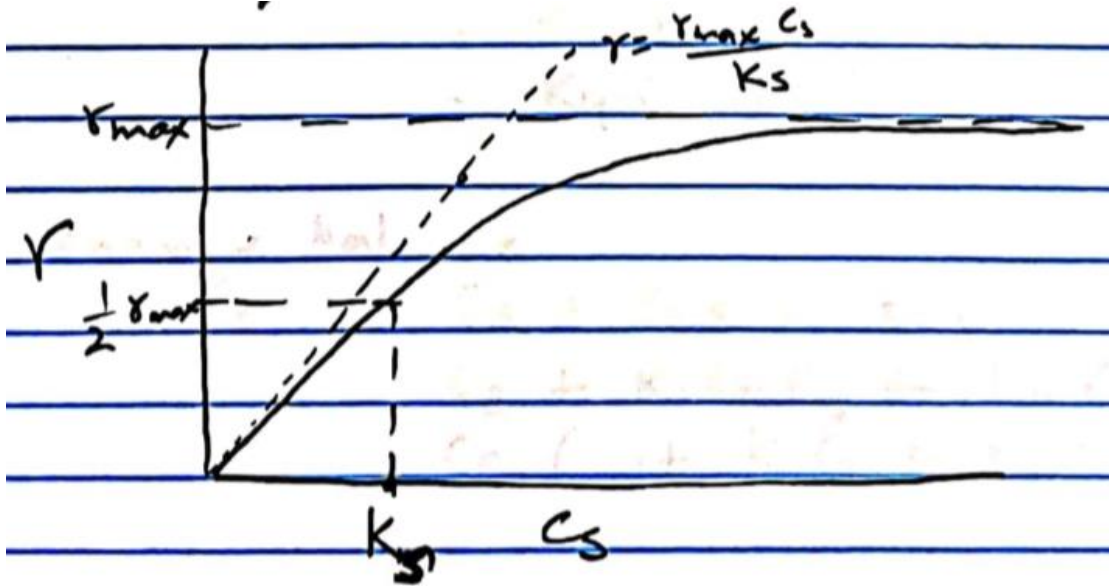
The parameters r_{max} and K_S are determined using Bork-Lineweaver plot.

Rearrange (78) to write

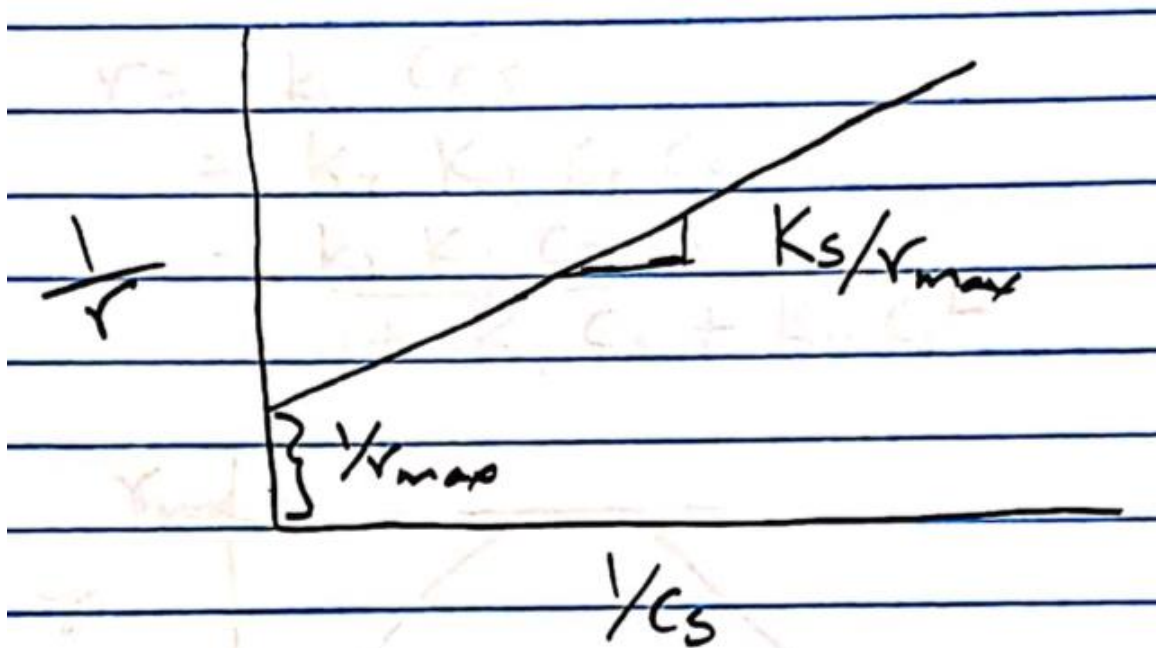
$$\frac{1}{r} = \frac{K_s}{r_{max}} \frac{1}{C_s} + \frac{1}{r_{max}} \quad (79)$$

Plot of $\frac{1}{r}$ vs $\frac{1}{C_s}$ is a straight line with slope $\frac{K_s}{r_{max}}$ and intercept $\frac{1}{r_{max}}$

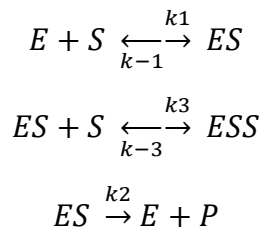
Plot eq (78) $\rightarrow$ Michaelis-Menton Curve



Plot eq (79) $\rightarrow$ Burk-Lineweaver plot



Now consider substrate inhibition such that



Assume:

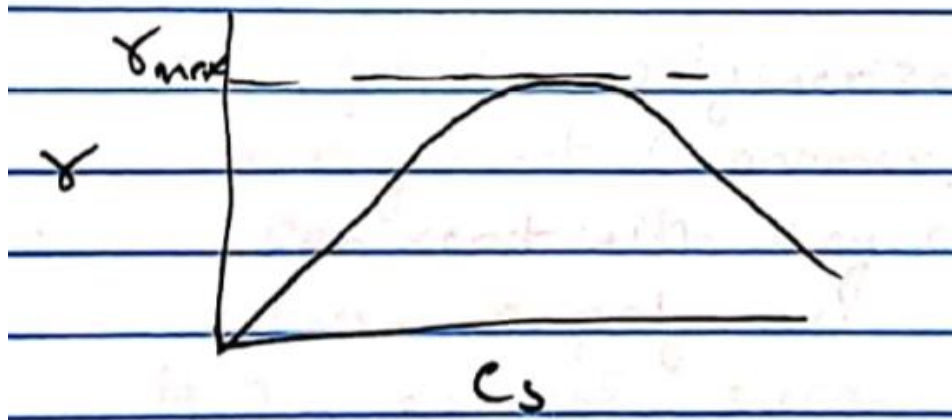
$$C_{ESS} = K_3 C_{ES} C_S = K_1 K_3 C_E C_S C_S = K_4 C_E C_S^2 \quad (80)$$

Enzyme balance

$$C_{E,0} = C_E + C_{ES} + C_{ESS} = C_E + K_1 C_E C_S + K_4 C_E C_S^2 = C_E (1 + K_1 C_S + K_4 C_S^2) \quad (81)$$

The rate of production:

$$r = K_2 C_{ES} = K_2 K_1 C_E C_S = \frac{K_2 K_1 C_{E,0} C_S}{1 + K_1 C_S + K_4 C_S^2} \quad (82)$$



$$r_{max} = \frac{K_2 K_1 C_{E,0}}{K_1 + 2K_4^{1/2}} \quad (83)$$

****3. Polymerization Kinetics**

*Polymers are long chain molecules formed by chemical bonding of a large number of small molecules called monomers.

*The chemical reaction transforming the monomers to polymers is called polymerization.

*A polymer molecule may feature several thousand monomer elements, with molecular weight of the order of a million.

*The mechanism of polymerization falls into two categories

(1) Step growth polymerization

*Any unit (monomer, dimer or n-mer can react with any other unit to form a polymer)

*In general, such process can be denoted as $P_n + P_j \rightarrow P_{n+j}$

(2) Chain growth polymerization

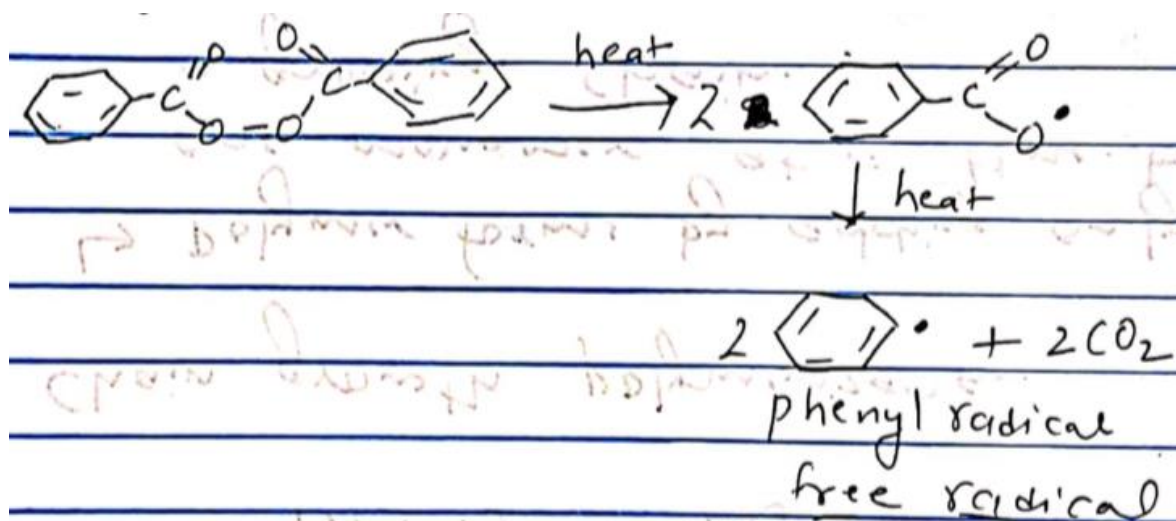
*Polymer forms by adding only one monomer at a time to the growing chain

*There are three major steps in chain growth polymerization

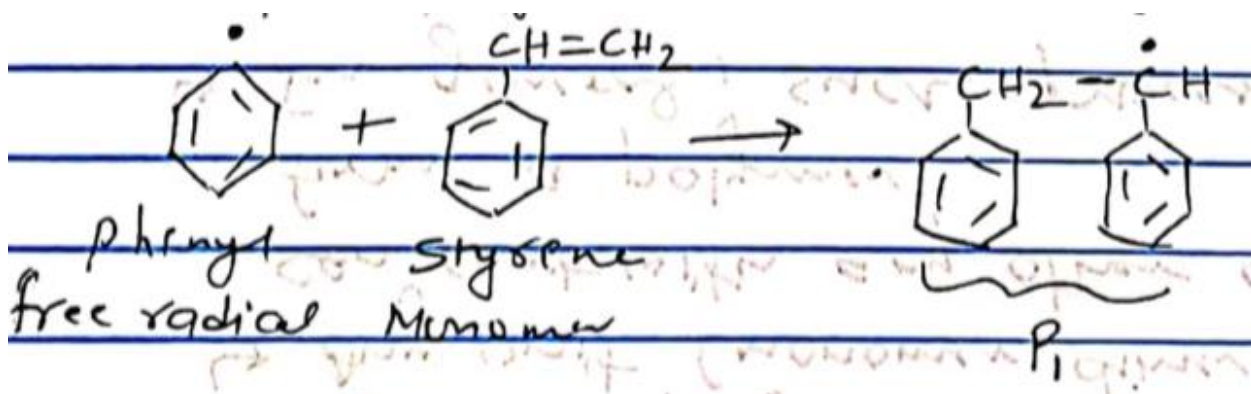
1. Initiation (free radical formation)



*Initiation is a mechanism to activate monomer using either a radical or ions.

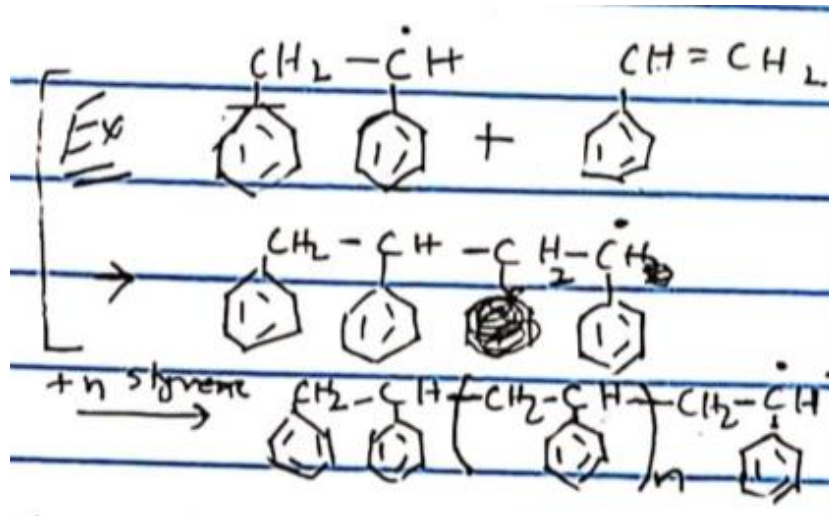


*Add benzoyl peroxide to styrene and heat

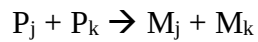


*Another mechanism to activate monomer is to use ions. Eg: $C_2H_5^- Li^+$

2. Propagation: $P_j + M \rightarrow P_{j+1}$



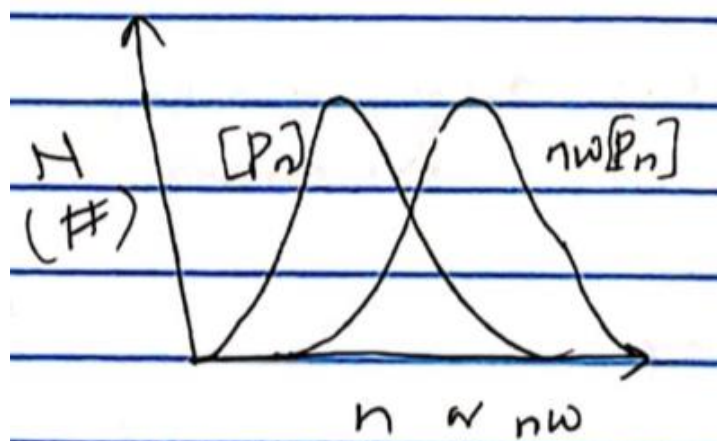
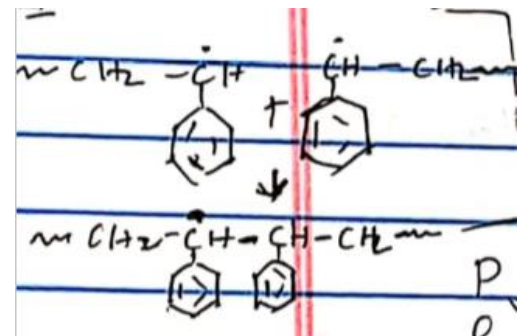
3. Termination:



* P_j is a growing free-radical chain of length j .

M_j is an inactive polymer of length j .

*The reaction results in a contribution of molecular weights.



*Let $[P_j]$ be the number of active polymers of length j . $[M_j]$ be the number of inactive polymer. $[M]$ be the number of monomers.

*The number average molecular weight. W is the molecular weight of monomer.

$$M_n = \frac{W \sum_j j [P_j]}{\sum_j [P_j]} \quad (84)$$

*Weight averaged molecular weight

$$M_w = \frac{\sum_j (jw)(jw[P_j])}{\sum_j (jw[P_j])} = \frac{w \sum_j j^2 [P_j]}{\sum_j j [P_j]} \quad (85)$$

$$\text{polydispersity} \equiv \frac{M_w}{M_n} = \frac{\sum_j j^2 [P_j] \sum_j [P_j]}{(\sum_j j [P_j])^2} = \frac{\langle j^2 \rangle}{\langle j \rangle} \quad (87)$$

*Dynamic light scattering can be used to measure M_n , M_w .

*Analysis of chain polymerization reaction with very fast initiation (After w. Harmon Ray and Lawrence (1976) in chemical reactor theory)

*Since radicals are very reactive species, the initiation reaction is very fast such that it can assume equilibrium.

Consider propagation of free radical chain: $P_n + M \rightarrow P_{n+1}$, $n = 1, 2, \dots$

The rate constant K_p is independent of n (Flory's Approximation)

The mole balance for species P_n & M can be written as

$$\frac{d[P_1]}{dt} = -k_p [M][P_1] \quad , [P_1](0) = [P_{1,0}] \quad (88)$$

$$\frac{d[P_n]}{dt} = -k_p [M]([P_n] - [P_{n-1}]) \quad , [P_n](0) = 0 \quad (89)$$

$$\frac{d[M]}{dt} = -k_p [M] \sum_{n=1}^{\infty} [P_n] \quad , [M](0) = [M_0] \quad (90)$$

Since $[P_{\infty}] \approx 0$, (88), (89) would give

$$\frac{d \sum_{i=1}^{\infty} [P_i]}{dt} = 0 \quad (91)$$

Let $[P] = \sum_{i=1}^{\infty} [P_i] \equiv \text{constant}$.

Also $[P] = \sum_{i=1}^{\infty} [P_i](t=0) = [P_{1,o}]$ (92)

Substituting (92) in (90)

$$\begin{aligned} \frac{d[M]}{dt} &= -k_p[M][P_{1,o}] \\ [M] &= [M_o]e^{-k_p[P_{1,o}]t} \end{aligned} \quad (93)$$

Note: k_p is a bimolecular rate constant with units $\text{vol} / (\#.\text{time})$

We can define a dimensionless time as

$$\tau = k_p \int_0^t [M](t') dt' \text{ or } d\tau = k_p [M] dt \quad (94)$$

Substitute (94) in (88) and (89)

$$\frac{d[P_1]}{d\tau} = -[P_1], \quad [P_1](0) = [P_{1,o}] \quad (95)$$

$$\frac{d[P_n]}{d\tau} = -([P_n] - [P_{n-1}]), \quad [P_n](0) = 0 \quad n > 1 \quad (96)$$

These (95) and (96) are infinite number of differential equations.

These infinite set of simultaneous ODEs can be solved using moment generating methods.

Let $G(s, \tau) = \sum_{n=1}^{\infty} s^n [P_n](\tau)$, $G(1, \tau) = [P]$ (97)

By differentiating $G(s, \tau)$ w.r.t s , we have means to generate moments of $[P_n]$

$$\frac{\partial G(1, \tau)}{\partial s} = \sum_{n=1}^{\infty} n [P_n](\tau) \quad (98)$$

Such that

$$\frac{\partial G(1, \tau)}{\partial s} = \sum_{n=1}^{\infty} n [P_n] = \mu_1 - \text{first moment} \quad (99)$$

$$\frac{\partial^2 G(1, \tau)}{\partial s^2} = \sum_{n=1}^{\infty} n^2 [P_n] - n [P_n] = \mu_1 - \mu_2 - \text{second moment} \quad (100)$$

Multiply (96) by s^n and sum over all n

$$\frac{\partial}{\partial \tau} \sum_{n=1}^{\infty} s^n [P_n] = \sum_{n=1}^{\infty} [P_{n-1}] - \sum_{n=1}^{\infty} [P_n]$$

Replace n by n+1

$$= \sum_{n=1}^{\infty} s^{n+1} [P_n] - s^n [P_n]$$

$$\rightarrow \frac{\partial G(s, \tau)}{\partial \tau} = (s - 1)G(s, \tau) \quad (101)$$

Initial condition: $G(s, 0) = s[P_{1,0}]$ (102)

Integrate (101)

$$G(s, \tau) = s[P_{1,0}]e^{(s-1)\tau} = s[P_{1,0}]e^{-\tau} \sum_{n=0}^{\infty} \frac{s^n \tau^n}{n!} = [P_{1,0}]e^{-\tau} \sum_{n=0}^{\infty} \frac{s^{n+1} \tau^n}{n!}$$

$$= [P_{1,0}]e^{-\tau} \sum_{n=1}^{\infty} \frac{s^n \tau^{n-1}}{(n-1)!} = \sum_{n=1}^{\infty} s^n \left[\frac{[P_{1,0}]e^{-\tau} \tau^{n-1}}{(n-1)!} \right] \quad (103)$$

Compare (103) with (97)

$$[P_n](\tau) = \frac{[P_{1,0}]e^{-\tau} \tau^{n-1}}{(n-1)!} \quad (104)$$

The number average molecular weight is give by (84)

$$M_n = \frac{w \sum_{n=1}^{\infty} n[P_n]}{\sum_{n=1}^{\infty} [P_n]} = \frac{w}{[P_{1,0}]} \frac{\partial G(1, \tau)}{\partial s} = \frac{w}{[P_{1,0}]} ([P_{1,0}] + \tau[P_{1,0}]) = w(1 + \tau) \quad (105)$$

The weight averaged molecular weight is given by (85)

$$M_w = w \frac{\sum_{n=1}^{\infty} n^2 [P_n]}{\sum_{n=1}^{\infty} n [P_n]} = w \frac{\mu_2}{\mu_1} = \frac{w}{[P_{1,0}](1 + \tau)} \left[\frac{\partial^2 G(1, \tau)}{\partial s^2} + \mu_1 \right]$$

$$= \frac{w}{[P_{1,0}](1 + \tau)} \left[[P_{1,0}]\{(1 + \tau)^2 + \tau\} \right] = w \frac{(1 + \tau)^2 + \tau}{1 + \tau} \quad (106)$$

Polydispersity is the given as:

$$\frac{M_w}{M_n} = 1 + \frac{\tau}{(1 + \tau)^2} \quad (107)$$

Similarly, we can obtain chain length distribution for step growth mechanism.

****Lumping in Reaction Analysis**

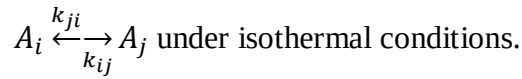
*When number of species is very large in a reaction system, it is important to seek reduction of dimensionality by “lumping” species together.

*Ex: In petrochemical industry, there exists a PONA analysis to represent four different lumps known as paraffins, olefins, naphthene, and aromatics.

*For a collection of species in a reaction system to be lumpable, one would expect variation of the composition of lump has limited effect on reaction rate.

*Wei (Princeton) and Kou(1969) developed a scheme to identify lumps of species in a large reaction network.

*Consider n species $A_1, A_2, \dots, A_n$ participating in the following reaction



*Mole balance for species A_j can be written as

$$\frac{dC_j}{dt} = k_{j1}C_1 + k_{j2}C_2 + \dots + k_{j,j-1}C_{j-1} - \sum_{i=1, j}^n k_{ij}C_j + k_{j,j+1}C_j + \dots + k_{jn}C_n, j = 1, 2, \dots, n \quad (108)$$

Alternatively, we may write in vector form

$$\underline{C} = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{bmatrix}, \underline{K} = \begin{bmatrix} -\sum_{i=1}^n k_{i1} & k_{12} & \dots & k_{1n} \\ k_{21} & -\sum_{i=1,2}^n k_{i2} & \dots & k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k_{n1} & k_{n2} & \dots & -\sum_{i=1}^{n-1} k_{in} \end{bmatrix}$$

So that

$$\frac{d\underline{C}}{dt} = \underline{K}\underline{C} \quad (109)$$

Lumping of reaction species implies species concentration be defined by $\widehat{\underline{C}}_1, \widehat{\underline{C}}_2, \dots, \widehat{\underline{C}}_p$ lumps where each vector is a lump of species.

